The Prop. 4 sign bias shrinks like $1/\sqrt{p}$

`research/smith_sign_scaling_20261016` · 2026-10-16 · protocol `PROTOCOL.md` frozen at `5fd6133d0` before any data

1. Question

At $p = 7, 11$ the Prop. 4 pass rate of Smith's construction follows $N_1/(2p+2)$ with $N_1 = \#H(\mathbb{F}_p)$ (`smith_serre_sign_20261016`), i.e. $r = 1/2 + \beta a_1/(2p+2)$ with $\beta \approx 1$. If β stays bounded, the bias at a typical curve ($|a_1| \sim \sqrt{p}$) is $O(1/\sqrt{p})$ and Smith's fair coin (Hypothesis 3) holds asymptotically; if β grows like $\sqrt{p}$, the bias persists at every size. This study measures β on 10^6 random curves at each of $p = 101, 1009, 10007$.

2. Verdict (frozen wording)

H3: scaling exponent γ of β_p in p (WLS on $\ln\beta$ vs $\ln p$)	$\gamma = 0.019$, 95% CI [−0.014, 0.051] → **bias shrinks like 1/√p (β bounded)** (rule: upper end < 0.25; persistence predicts 0.5)

p	squarefree curves	non-degenerate Prop. 2 passes	degenerate	α (control: 0.5 ± 0.02)	β [95% CI, CR1]	H1′: CI contains 1
101	990071	491354	4681	0.5007	1.019 [0.991, 1.047]	`true`
1009	999030	496390	523	0.4993	1.092 [1.003, 1.18]	`false`
10007	999900	500199	52	0.5005	0.942 [0.665, 1.219]	`true`

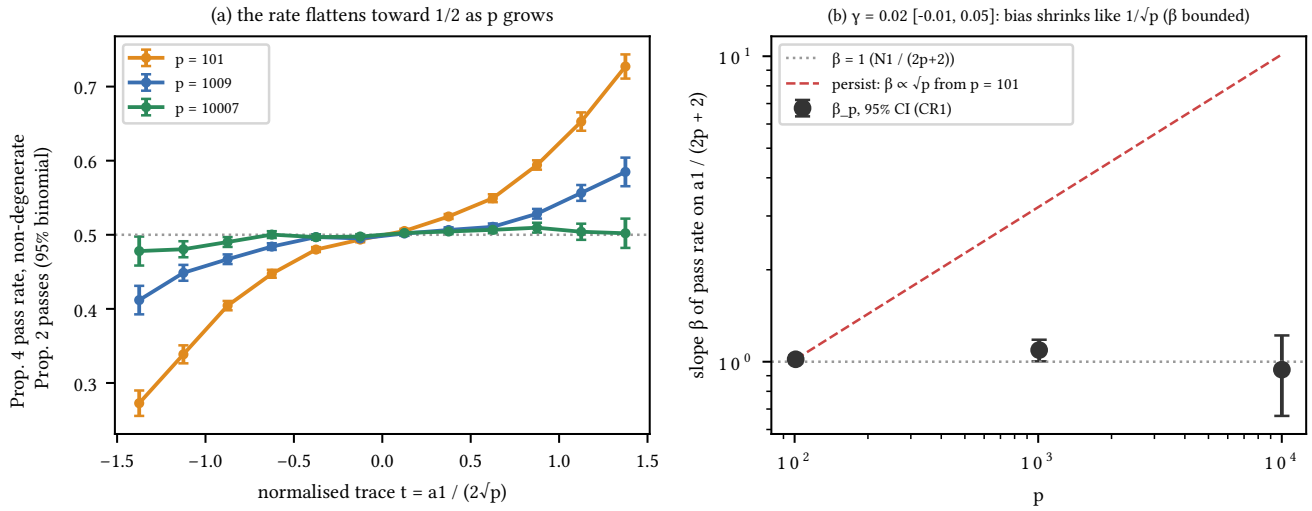


Figure 1: (a) Prop. 4 pass rate of non-degenerate Prop. 2 passes against the normalised trace $t = a_1/(2\sqrt{p})$ (bins with ≥ 2,000 subgroups, 95% binomial intervals): the dependence flattens toward 1/2 as p grows. (b) Fitted slope β_p on $a_1/(2p+2)$ with 95% curve-clustered intervals, against $\beta = 1$ and the persistence alternative $\beta \propto \sqrt{p}$. Source `figures/plot.py`, data `data/scaling_p*.json`, `data/verdict.json`.

3. Findings

- **Measured, frozen rule.** β_p does not grow with p: $\gamma = 0.019$ [−0.014, 0.051], far from the 0.5 that persistence predicts. The Prop. 4 sign bias is $\beta a_1/(2p+2)$ with β bounded, so at a typical curve it is $O(1/\sqrt{p})$; Smith's fair-coin Hypothesis 3 is recovered as p grows, and the small-p clustering of `full_success` within L-classes is a small-prime effect.
- **Measured.** β stays close to 1 from $p = 7$ to $p = 10007$ (1.02, 1.09, 0.94 at $p = 101, 1009, 10007$): the pass rate is close to $\#H(\mathbb{F}_p)/(2p+2)$ throughout. At $p = 1009$ the interval [1.003, 1.18] just excludes 1; the three primes do not show a trend in that deviation.
- **Measured.** Degenerate Prop. 2 passes (Prop. 4 fails on both twists) fall from 4681 at $p = 101$ to 52 at $p = 10007$ per about $5 \cdot 10^5$ passes, consistent with a small-p construction failure.
- **Interpretation (labelled, not tested).** $r = \#H(\mathbb{F}_p)/(2p+2) = \#X_+(\mathbb{F}_p)/(\#X_+(\mathbb{F}_p) + \#X_-(\mathbb{F}_p))$, where $X_\pm$ are the two candidate quotient quartics: the outcome behaves as if a rational point were chosen uniformly on $X_+ \cup X_-$. No mechanism in Smith's construction is identified here that would explain this.

4. Validation

All three primes valid: intercept within 0.5 ± 0.02 (0.5007, 0.4993, 0.5005); native point count equal to `smith_prop24::h_point_count` on the first 1,000 squarefree curves (0, 0, 0 mismatches); no curve with more Prop. 4 passes than Prop. 2 passes. Standard errors are clustered by curve (CR1).

5. Scope

Three primes up to 10007, random degree-8 models. "Shrinks like $1/\sqrt{p}$" is the trend over this range, not a theorem. The residual quarter of the within-class clustering at $p = 7, 11$ stays open; a post hoc check (protocol, "Also recorded") found that the number of Prop. 2 passes adds nothing and N_3 at most a few points beyond a random label. The mechanism behind the $\#X$-proportional rule is not tested.

6. Reproduce

```
cargo build --release --example smith_sign_scaling
B=target/release/examples/smith_sign_scaling; D=research/smith_sign_scaling_20261016/data
for p in 101 1009 10007; do $B run --p $p --n 1000000 --out $D; done   # 50 s, 82 s, 225 s on 2 vCPU
$B verdict --dir $D 101 1009 10007
python research/smith_sign_scaling_20261016/figures/plot.py            # layout only
```